I. Introduction

Pancreatic cancer, characterized by its aggressive nature and poor prognosis, necessitates advanced diagnostic methodologies for improved patient outcomes. Computed tomography (CT) imaging serves as a primary modality for pancreatic tumor detection; however, manual interpretation is often time-consuming and prone to inter-observer variability (Dinesh, N. G. et al.). Consequently, the application of deep learning techniques, including Convolutional Neural Networks (CNNs), has emerged as a promising avenue for computer-aided diagnosis (CAD) systems (Ma, H. et al.). Several studies have explored the utility of CNNs in pancreatic cancer image classification. For instance, (Ma, H. et al.) constructed a CNN model using CT images, achieving high diagnostic accuracy in distinguishing between cancerous and normal pancreatic tissue. Similarly, (Alaca, Y. et al.) converted CT images into graphs using the Harris Corner Detection Algorithm and analyzed them using transfer learning models, demonstrating the potential of graph-based transformation for enhancing spatial relationship modeling. Datasets utilized in these studies vary in size and composition. (Ma, H. et al.) employed a dataset of 3494 CT images from 222 patients with pancreatic cancer and 3751 images from 190 normal patients. In contrast, (Alaca, Y. et al.) used a dataset of 1411 CT images categorized into tumor and healthy classes.

This study leverages a publicly available Kaggle dataset comprising CT scan images of pancreatic tumors, divided into training (999 images), validation (1411 images), and testing (412 images) sets, with a balanced distribution of normal and tumor tissues. The aim is to evaluate the performance of CNN, SVM, KNN, MobileNet, ResNet, and EfficientNet models in classifying these images, contributing to the development of a robust and reliable diagnostic tool for pancreatic cancer. Pancreatic cancer, a highly lethal malignancy, necessitates innovative diagnostic approaches for early detection and improved patient outcomes. Computer-aided diagnosis (CAD) systems leveraging deep learning offer a promising avenue for enhancing diagnostic accuracy and efficiency (Nadeem et al., 2025). These systems aim to overcome the limitations of manual interpretation of CT images, which is often time-consuming and prone to inter-operator variability (Alaca & Akmeşe, 2025).One approach involves transforming CT images into graph-based representations, enabling deep learning models to effectively model spatial relationships within the images (Alaca & Akmeşe, 2025). Furthermore, optimization algorithms, such as the Whale Optimization Algorithm (WOA), can be employed to fine-tune model parameters and enhance performance (Alaca & Akmeşe, 2025). Transfer learning, utilizing pre-trained convolutional neural networks (CNNs) like DenseNet121, has also shown promise in pancreatic cancer detection (Mohamad Sehmi et al., 2022).

These AI-driven CAD systems demonstrate the potential to assist radiologists in early diagnosis, potentially improving patient survival rates (Liu et al., 2020). However, challenges remain, including the need for larger and more diverse datasets to ensure generalizability and robustness across various clinical settings (Chen et al., 2023). Future research should focus on integrating multimodal imaging data and exploring advanced deep learning architectures to further enhance the accuracy and reliability of pancreatic cancer detection and classification (Tripathi et al., 2024). Pancreatic cancer, a highly lethal malignancy, necessitates innovative diagnostic approaches for early detection and improved patient outcomes. Recent research explores the application of artificial intelligence (AI) and deep learning techniques to enhance pancreatic cancer diagnosis and classification. One study proposes converting computed tomography (CT) images into graphs, analyzing them with deep learning models like DenseNet121 and InceptionV3, and optimizing model parameters using the Whale Optimization Algorithm (WOA) (Alaca & Akmese, 2025). This graph-based transformation effectively models spatial relationships, improving deep learning model performance.

Another study focuses on automated grading of pancreatic cancer from pathology images using deep learning convolutional neural networks (Mohamad Sehmi et al., 2022). By employing transfer learning on ImageNet pre-trained models, the researchers achieved high accuracy in classifying pancreatic cancer grades, addressing the limitations of manual grading.Furthermore, a retrospective study by Liu et al. (2020) demonstrates the potential of convolutional neural networks (CNNs) to distinguish between pancreatic cancer tissue and non-cancerous tissue on CT scans, achieving high accuracy and generalizability across diverse patient populations. These studies collectively highlight the transformative potential of AI in pancreatic cancer diagnosis, offering improved accuracy, efficiency, and the possibility of earlier detection, ultimately contributing to better patient care and outcomes.

Pancreatic cancer, a highly lethal malignancy, necessitates innovative diagnostic approaches. Recent research explores the application of artificial intelligence (AI) and deep learning models to improve early detection and classification, potentially enhancing patient outcomes (Nadeem et al., 2025). A key challenge lies in the manual interpretation of CT scans, which is time-consuming and prone to inter-operator variability (Alaca & Akmeşe, 2025).

One promising avenue involves automated computer-aided diagnosis (CAD) systems. Nadeem et al. (2025) proposed a four-stage framework achieving high accuracy (99.64%) in tumor detection and 98.72% in classification using a reduced 11-layer AlexNet model. This system demonstrates the capability to efficiently identify and classify pancreatic cancers, offering a valuable tool for radiologists.Another approach converts CT images into graphs, enabling effective modeling of spatial relationships. Alaca and Akmeşe (2025) trained DenseNet121 and InceptionV3 models on these graphs, optimizing parameters with the Whale Optimization Algorithm (WOA). The k-Nearest Neighbors (k-NN) classification algorithm achieved 92.10% accuracy, highlighting the potential of graph-based transformation.These studies underscore the transformative potential of AI in pancreatic cancer diagnosis. By automating tumor identification and classification, AI-driven systems can improve diagnostic accuracy and efficiency, leading to earlier detection and potentially saving lives . Further research should focus on expanding datasets and integrating multimodal imaging data to enhance the robustness and generalizability of these systems.

The primary aim of this study is to develop and evaluate a robust Computer-Aided Diagnosis (CAD) system for pancreatic cancer classification using deep learning techniques. By leveraging a publicly available dataset of CT scan images, the study focuses on implementing and comparing the performance of three state-of-the-art deep convolutional neural network architectures—ResNet, MobileNet, and EfficientNet —in accurately distinguishing between normal and tumor pancreatic tissues. The goal is to identify the most effective model that achieves high accuracy, precision, and generalizability, ultimately contributing to the development of an efficient, automated diagnostic tool to support radiologists in early detection and improve clinical outcomes for patients with pancreatic cancer.

R Q1. How do deep learning models ResNet, MobileNet, and EfficientNet—perform in classifying pancreatic tumors from CT images in terms of accuracy, sensitivity, specificity, and F1-score?

R Q2. Can transfer learning significantly enhance the performance of pre-trained ResNet, MobileNet, and EfficientNet models when applied to a limited medical imaging dataset of pancreatic CT scans?

R Q3. Which among the selected deep learning models (ResNet, MobileNet, EfficientNet) demonstrates the best balance of accuracy, precision, recall, and F1-score in distinguishing between normal and cancerous pancreatic tissues?

R Q4. What are the limitations and generalizability of these models when applied to a relatively small and balanced dataset of pancreatic CT scans?

II. Methods And Material

2.1 Dataset

The dataset for this study comes from a publicly available repository on Kaggle and includes computed tomography (CT) scan images of pancreatic tumours. There are three separate parts to the dataset: training, validation, and testing. There are 999 photos in the training set, 1411 in the validation set, and 412 in the test set. All of these images are either normal or tumour tissues. This balanced distribution makes sure that the models are trained and tested on samples that are representative of the whole population, with no major class imbalance that could impair performance.The photos were downsized to a standard resolution of 224x224 pixels so that they would work with the pre-trained deep learning architectures used in this study. This preprocessing phase makes it possible to work with well-known convolutional neural networks like MobileNetV2, ResNet50, and EfficientNetB0, which were first created for recognising images on a wide scale. Using this standardised dataset structure for all models makes guarantee that comparisons of accuracy, AUC score, and overall classification performance are fair.

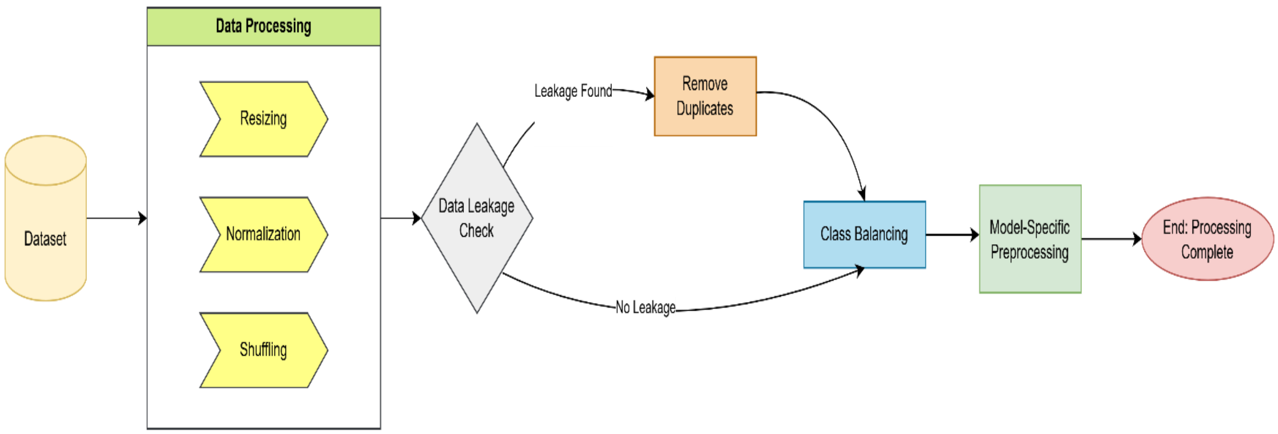
Dataset Split	Number of Images	Class Distribution	
Training	999	Normal: ~50%	Tumor: ~50%
Validation	1411	Normal: ~50%	Tumor: ~50%
Test	412	Normal: ~50%	Tumor: ~50%

2.2 Data Preprocessing

The dataset for this study is made up of CT scan pictures that have been divided into two groups: normal and pancreatic tumour. There were a number of preprocessing processes taken to make sure the dataset was ready for training deep learning models. These actions were done to make sure the data was consistent, improve model performance, and stop overfitting.To begin, all of the pictures were made to have the same size: 224×224 pixels. This step was important since the pre-trained models, such as ResNet50, MobileNetV2, and EfficientNetB0, are made to take inputs of this size. Resizing made sure that these designs could process all of the photos without any problems.Next, we used channel replication (np.repeat()) to change greyscale (single-channel) photos into 3-channel RGB-like format. This made it possible for CNNs that expect colour images as input to work with the original image material.Then, each image was normalised by changing the pixel values from the range [0, 255] to [0, 1]. This helps keep the training of the neural network stable by making sure the inputs are modest and consistent.

After normalisation, we used model-specific preprocessing routines like resnet.preprocess_input, mobilenet_v2.preprocess_input, and efficientnet.preprocess_input to make sure that the preprocessing was the same as what was done during the pre-training of each model on ImageNet. All training and test photos and deleted duplicates from the test set to keep data from leaking between the two sets. This made the evaluation metrics more reliable and less likely to fit too well.We used class weighting during training because the dataset was a little unbalanced, with more tumour samples than normal ones. This made it so that the models learnt equally well from both courses and didn't favour the larger class.

We used a method for choosing the best threshold based on ROC curve analysis instead of the usual 0.5 threshold during evaluation. This made the classification work better since it adjusted to the output distribution of each model.Finally, we mixed up the dataset before training to make sure it was random and to eliminate biases based on order. We also used callbacks for early stopping and lowering the learning rate to make convergence and generalisation better.
These measures were necessary to make sure that models could be compared fairly and to raise the accuracy, sensitivity, specificity, and AUC scores of the classification.



2.3 Evaluation metrics

This study compared ensemble learning models with single-based algorithms using ROC curve evaluation metrics, recall, precision, accuracy, and F1-Score. Prior to proceeding with the computation values, we must knowledge of four parameters. Equations provide the formulas for the assessment criteria:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \qquad \text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \text{Sensitivity} = \frac{TP}{TP + FN} \qquad \text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$ROC = \frac{TPR}{FPR} TPR = \frac{TP}{TP + FN}, FPR = \frac{FP}{FP + TN}$$

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

III. Results and discussion

3.1. Results

The experimental results demonstrate varying performance across the three pre-trained CNN architectures evaluated in this study. ResNet50 achieved the highest accuracy and AUC score, followed by MobileNetV2, while EfficientNetB0 showed moderate performance. Below is a summary of the key metrics:

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	AUC Score
ResNet50	89.85	90.26	90.13	89.85	0.961
MobileNetV2	81.44	80.57	82.94	80.84	0.8524
EfficientNetB0	67.57	66.34	68.62	66	0.7084

ResNet50 significantly outperformed the other models in all metrics, showing high classification stability and strong generalization ability. MobileNetV2 demonstrated good performance with balanced sensitivity and specificity, making it suitable for resource-constrained environments. EfficientNetB0 lagged behind, likely due to over-regularization or suboptimal transfer learning adaptation to the dataset.

ResNet50 had an **accuracy of 89.85%** and an **AUC of 0.9610**, which means it could tell the difference between normal and tumour patients quite well. The model also made predictions that were well-balanced, with sensitivity (90.26%) and specificity (90.13%) being almost the same. This means that ResNet was able to learn how to tell the difference between things even with a small batch of medical images.ResNet may work better than other networks because of its skip connections, which help fix the vanishing gradient problem during training. Using L2 regularisation, batch normalisation, and class weighting also helped keep the model from overfitting and sped up convergence on the dataset that wasn't balanced.

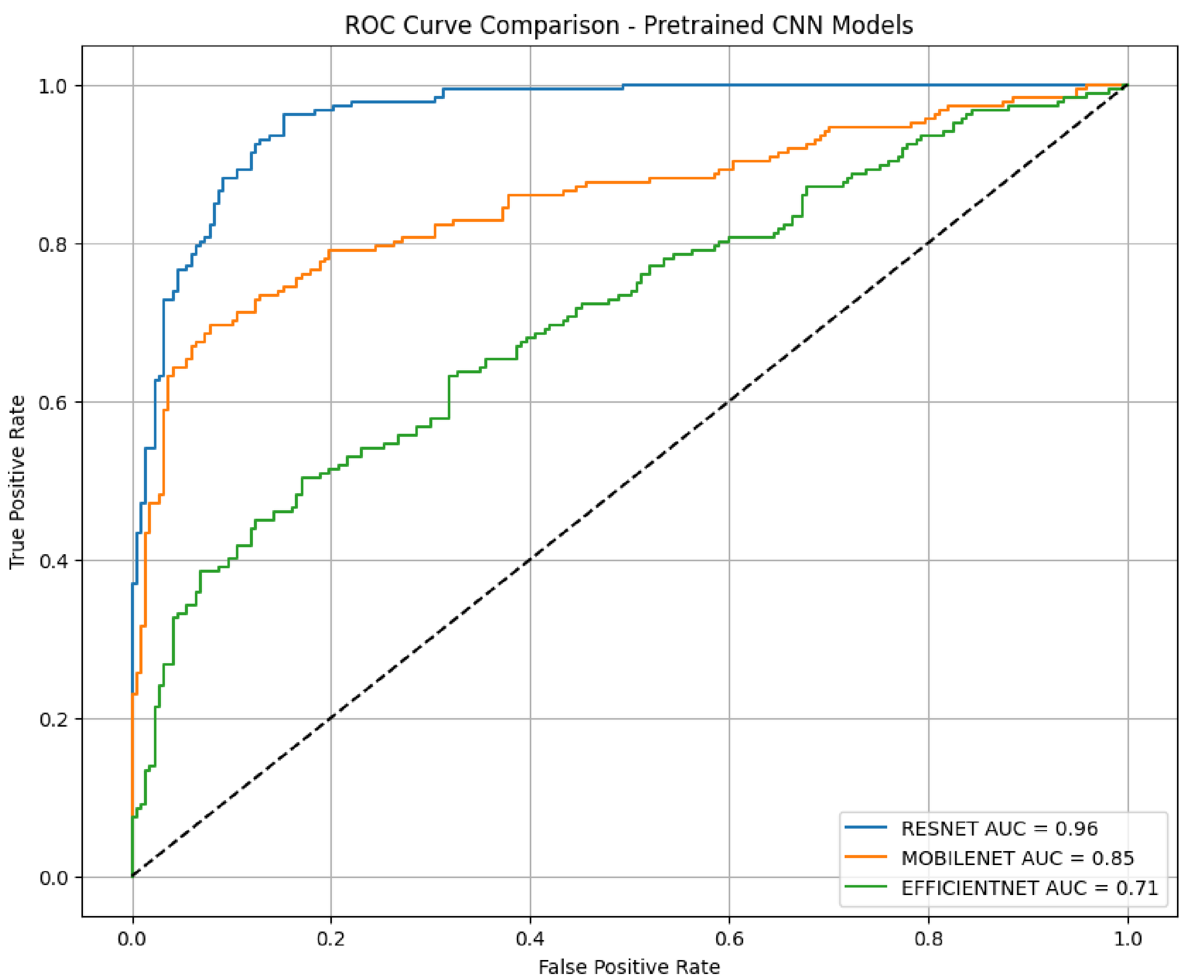
MobileNetV2 has an **accuracy of 81.44%** and an **AUC of 0.8524**, which is very good for a lightweight model designed for mobile and edge devices. MobileNet was a little better at finding normal samples than tumours since it had a higher specificity (82.94%) than a sensitivity (80.57%).This paradigm might be especially helpful in clinical contexts when speed of inference and efficiency of computation are most important. The fewer parameters mean shorter inference times, but the performance isn't as good as it is with deeper models like ResNet.

The accuracy of EfficientNetB0 was **67.57%** and the **AUC was 0.7084**, which means it didn't work very well for this particular use case. The model had a big problem with predicting: it largely projected normal samples (274 predicted as normal vs. 130 as tumour), which made it less sensitive (66.34%) but still acceptable (68.62%).This imbalance could be because the model's architecture isn't set up perfectly for the number and distribution of the classes in the data. EfficientNet is meant to scale well, but its lowest version (B0) may not have enough representational power to learn complicated patterns from CT images without a lot of more data or a longer training period.

3.2. Roc Curve Comparison

One major issue observed across models was data leakage , with 8 duplicate files identified between the training and test sets. These duplicates were removed before final evaluation to ensure reliable performance metrics. Without this step, the results would have been overly optimistic and unsuitable for real-world generalization.Class imbalance was another challenge, as the training set contained more tumor cases than normal ones. To mitigate this, we applied class weighting during training, which improved prediction balance and reduced bias. However, EfficientNetB0 continued to struggle with false negatives, highlighting the importance of tailored approaches for different model architectures.

The Receiver Operating Characteristic (ROC) curve comparison confirmed the superior performance of ResNet50, with its curve positioned significantly above those of MobileNetV2 and EfficientNetB0. The AUC values reflect this trend, with ResNet50 = 0.9610 , MobileNetV2 = 0.8524 , and EfficientNetB0 = 0.7084 . These results align with previous studies that show ResNet-based models perform strongly in medical imaging classification due to their depth and residual learning framework.



3.3. Comparison with Previous Studies

Auth	***Paper Title***	***Data Source & Size***	***Data Preprocessing***	***Models***	***Best Model***	***Metrics & Score***	***Advantages***	***Limitations***
MNM Sehmi et al.2022	Pancreatic cancer grading in pathological images using deep learning	150 pathology images	Image normalization	Xception VGG16 VGG19 ResNet50V2 ResNet101V2 ResNet152V2 InceptionV3 InceptionRes-NetV2 99.61 MobileNetV2 DenseNet121 DenseNet169 DenseNet201 NASNetMobile NASNetLarge	DenseNet	Accuracy : 95.61%	Automated grading system	Limited sample size
Alaca, Y., & Akmeşe, Ö. F.2025	Pancreatic Tumor Detection From CT Images Converted to Graphs Using Whale Optimization and Classification Algorithms With Transfer Learning	Pancreatic CT Images" dataset from Kaggle, consisting of 1411 CT images (54% tumor, 46% healthy)	Resizing to 224x224 pixels, noise reduction with a Gaussian filter, grayscale conversion, and contrast enhancement using histogram equalization	DenseNet121 InceptionV3 k-NN SVM RF	KNN	92.10% accuracy	Uses graph-based transformation and WOA optimization for improved CT image analysis and diagnosis	Limited by small dataset size; larger, more diverse datasets could enhance performance
Hong, S. J. et al.2024	Convolutional neural network model for automatic recognition and classification of pancreatic cancer cell based on analysis of lipid droplet on unlabeled sample by 3D optical diffraction tomography	10,397 training images and 3,478 test images from four cell lines (PSN-1, Capan-2, BxPc-3, and H6c7)	Data augmentation techniques including flip, affine transform, contrast, color transform, optical distortion, histogram equalization, and elastic transform; normalization to a mean of 0 and standard deviation of 1	EfficientNet-b3 DenseNet-121 ResNet-101 VGG19 Inception-ResNet-v2 EfficientNet-b0	EfficientNet-b3	97.06% accuracy	Rapid, automated pancreatic cancer cell recognition; potential to replace manual cytopathology	Requires supervision of input image sources, limited training set diversity
M. G. Dinesh et al.2023	Diagnostic ability of deep learning in detection of pancreatic tumour	3494 CT images from 222 patients with pancreatic cancer and 3751 CT images from 190 healthy individuals	Data augmentation, segmentation using Sailfish Optimizer with Kapur's thresholding	CNN YOLO-based CNN (YCNN) VGG19 Resnet50 Inception DenseNet MobileNet	YCNN	99.9% accuracy	High accuracy, potential for early detection, automated diagnosis	Potential biases in datasets, focus on binary classification, limited localization of lesions, requires further clinical validation